KCE081BCT035(PREDICTING INSURANCE CLAIMS)

1. Target Variable Analysis (`claim_status`)

- Analyze distribution of insurance claims:
  - Claimed vs Not Claimed
- Plan resampling (SMOTE/undersampling) if imbalance exists

2. Demographic & Customer Analysis

- Analyze `customer_age`, `region_code`, `region_density`, `segment` vs claim status
- Identify age groups or segments with higher claim probability
- Compare urban vs rural (`region_density`) claim patterns

3. Vehicle Age & Model Analysis

- Analyze `vehicle_age` and `model` vs claim status
- Identify older or specific models with higher claim likelihood
- Examine correlation between vehicle age and claim probability

4. Vehicle Specifications & Performance

- Analyze numerical vehicle features:
  - `displacement`, `cylinder`, `gross_weight`, `turning_radius`, `length`, `width`, `max_power`, `max_torque`, `ncap_rating`
- Examine correlation of vehicle performance and safety ratings with claim occurrence

5. Safety & Technology Features

- Analyze features such as:
  - `airbags`, `is_esc`, `is_tpms`, `is_parking_sensors`, `is_brake_assist`, `is_power_steering`, etc.
- Identify which safety features reduce claim probability
- Compare vehicles with and without specific features

6. Fuel Type & Transmission Analysis

- Examine `fuel_type` and `transmission_type` vs claim status
- Identify patterns in claim occurrence based on vehicle configuration

7. Interaction & Combined Effects

- Analyze combined effects such as:
    - Vehicle age × Safety features
    - Customer age × Vehicle type
    - Region × Vehicle model
- Identify high-risk combinations for insurance claims

8. Feature Engineering Insights

- Engineer new features such as:
    - Vehicle age categories
    - Safety score (combining airbags, ESC, NCAP rating)
    - Vehicle size index (length × width × gross_weight)
    - Customer risk index (age × segment × region_density)